BASIC ADMINISTRATION

ROOT ACCOUNT AND SUDOERS

THE ROOT ACCOUNT

- Linux has a built-in account called 'root'
- The root account is the superuser account
  - This account has the privilege to manage the system
- The Debian system does not allow login directly as the root
  - By default a user account is created upon installation
  - The created user account can 'sudo' as the root to manage the system
  - Other accounts can also be a 'sudoer' by modifying the sudo list

THE 'SUDO' COMMAND

- The `sudo` command allows execution of commands for the root account
  - Usage:

```
sudo <command to be used>
```

  - Other accounts can be put in the 'sudoer' file by using the '`visudo`' command

THE '`SYSTEMCL`' COMMAND

- The systemctl command can be used to control the runlevel and service in the Linux system
- This has replaced the `initd` program used before to change runlevels and system startup
  - Linux now used systemd program to handle runlevels and startup

RUNNING AS ANOTHER USER

su [user]

- Switches to another user account.
- Will prompt for the password of the account to switch to
- If user is not specified, it defaults to switching to the root account

RUNNING AS ANOTHER USER

sudo [command]

- Runs the succeeding command as another user

Option	What it does
-u [user]	Run command as the specified user
-i	Simulate initial login (runs initialization files of target user)
-s [shell]	Run command in specified shell

- Note: To use the sudo command, a user must be included in the **/etc/sudoers** file
  - Option 1: Use '**visudo**' command to edit the file (Never do it directly from a text editor!)
  - Option 2: Add the user to the '**sudo**' group if it exists

THE SUDOERS FILE

user_list host_list = [(runas_list)] command_list

Ex: `administrator ALL = (root) ALL`

user_list	- specifies the user or users this line applies to, this can be usernames, groups (must be specified with %)
host_list	- specifies the host or hosts this line applies to, this can be hostnames, ip addresses
runas_list	- specifies the user that the user in the user_list can run as, this can be usernames, groups (must be specified with %)
command_list	- specifies the commands that the user_list can run Note: command format should be an absolute filename

SERVICES AND RUNLEVELS

SYSTEMCTL FOR RUNLEVELS

- Runlevels are called targets on the Linux system
- Targets or runlevels allows running of different programs upon start-up
- Targets in the Linux system are:

Runlevel	Target
0	poweroff.target
1	rescue.target
2, 3, 4	multi-user.target
5	graphical.target
6	reboot.target

SYSTEMCTL FOR RUNLEVELS

- Command for loading different targets

```
sudo systemctl isolate <target name>
```

SYSTEMCTL FOR SERVICES

- Services can be controlled by using systemctl
- Services are daemons or programs running in the background that provide a functionality
- To check the status of a service:

```
sudo systemctl status <name of service>
```

- To start a service

```
sudo systemctl start <name of service>
```

- To stop a service

```
sudo systemctl stop <name of service>
```

ADMINISTRATOR UTILITIES

TIMEDATECTL

- Date, time and time zone can be changed using the timedatectl command
- Timdatectl has the following commands

Command	Description
status	Shows the current date, time, time zone and other configuration on the system time.
list-timezones	List the available time zones that can be set on the system
set-timezone	Sets the time zone of the system
set-ntp	Turn on or off time synchronization on the system using NTP (Network Time Protocol)
set-time	Sets the time and date of the system.

TIMEDATECTL

- To turn on / off time synchronization

```
sudo timedatectl set-ntp on/off
```

- Change time zone

```
sudo timedatectl set-timezone <timezone>
```

- Change date

```
sudo timedatectl set-time <hh:mm>
```

- Change time

```
sudo timedatectl set-time <yyyy/mm/dd>
```

CRON

- The '`cron`' utility allows an administrator or user to setup a program or application to be executed by the system at a specific time
- The root account has automatic access to the '`cron`' utility but users has to be given permission by the root.
- The 'cron' utility configuration file is the '`/etc/crontab`' file
  - The administrator can edit the file by entering the date, time or day of the week and the program to be executed as specified by the time
  - Users can be given permission to enter programs or task to be executed in the '`/etc/cron.allow`' or '`/etc/cron.deny`'

BACKUP AND RESTORE

- The `tar` utility allows the user to pack files into one large file
  - It can also compress the whole archive
  - Came from the term "tape archive"
- Actually used to backup and restore files in the Unix/Linux system on a tape drive

tar <-option> <argument 1> <argument 2>

Options	Description
c -	Create archive
v -	Verbose
f -	The target or source is a file
x -	Extract file or files
z -	Compressed

NETWORK CONFIGURATION THRU NETPLAN

- Network configuration can be modified using a file inside `/etc/netplan`
- File name depends on the device
- File uses indentation to know which setting is being modified
  - Indentation should be strictly followed
- Example `netplan` configuration

```
network:
    ethernets:
        ens18:
            dhcp4: no
            addresses: [192.168.0.226/24]
            gateway4: 192.168.0.1
            nameservers:
              addresses: [192.168.0.1]
    version: 2
```

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